MIL-D-23000(WEP)
AMENDMENT-2
15 JUNE 1963
SUPERSEDING
AMENDMENT-1
15 FEBRUARY 1962

MILITARY SPECIFICATION

DECODER GROUP OA-3369(*)/APX-7

This amendment forms a part of Military Specification MIL-D-23000(WEP) dated 15 September 1961 and has been approved by the Bureau of Naval Weapons, Department of the Navy.

*Page 1, paragraph 1.2: Delete and substitute:

"1.2 Classification - Decoder Group OA-3369(*)/APX-7 shall be of one type and shall consist of the following items.

Items	Type Designation	Max. Allow. Size (Inches)	Max. Allow. Wt. (Lbs.)	Appl. Para.
Control-Indicator	C-3707(*)/APX-7	7-1/4 by 7-1/4 by 6	7	3.4.1
Decoder, Pulse	KY-384(*)/APX-7	15-1/4 by 6-1/2 by 5-3/4	9-1/2	3.4.2
Decoder, Video	KY-364(*)/APX	5-13/16 by 10-1/8 by 8-1/4	12	3.4.3
Mounting	MT-2648(*)/APX-7	15-1/4 by 2-1/4 by 6-1/8	1	3.4.4"

*Page 2, paragraph 2.1, under SPECIFICATIONS Military: Add:

"MIL-R-23094 Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for."

*Page 3, paragraph 3.2.2: Delete "28 1/2 " and substitute:

"29 1/2"

*Page 4: Add:

"3.2.3.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have a minimum of 250 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3."

*Page 5, paragraph 3.2.10(1): Delete "1.5 Amps" and substitute:

"2.8 Amps"

*Page 6, paragraph 3.3.1b, two parenthetical sentences: Delete and substitute:

"(Azimuth variable in width from 5° or less to 12° or more. Range variable in length from 12 microseconds or less to 250 microseconds or more)"

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*Page 6, paragraph 3.3.2.1, lines b and c: Delete and substitute:

"b. Amplitude 4 to 8 volts
c. Baseline +11.2 ±0.6 volts"

*Page 6, paragraph 3.3.2.2, lines a., b. and c: Delete and substitute:

"a. Duration 0.50 ±0.10 usec
b. Amplitude 5.6 ±0.75 volts
c. Baseline + 11.2 ±0.6 volts"

*Page 7, paragraph 3.3.2.3: Delete and substitute:

"3.3.2.3 Azimuth Synchro Voltage from Indicator -

(a) Frequency: 400 ±20 cps
(b) Phase Voltage:

Antenna Pos. Relative to Cursor

	0° or 180°	90° or 270°
Phase 1	22.0 ±3.0 VRMS*	0 to 0.2 VRMS*
Phase 2	0 to 0.2 VRMS*	22.0 ±3.0 VRMS*

(c) Reference Voltage:

26.0 ±1.5 VRMS

* Measured when loaded only by Decoder Group OA-3369/APX-7."

*Page 7, paragraph 3.3.2.4: Delete "20 ±5 volts Normal (10V Min)" and substitute:

"10 to 25 volts"

*Page 7, paragraph 3.3.3.1: Delete and substitute:

"3.3.3.1 Range-Azimuth Gate Signal - The gate output shall be a series of pulses, positive polarity, mixed with IFF video. The gate pulse average amplitude shall be adjustable over the range of 1.0 to 3.0 volts."

Page 8, paragraph 3,3,4,3: Delete "20°C" and substitute:

"-20°C"

*Page 9, paragraph 3.4.1.1: Delete "both units of the Decoder Group" and substitute:

"the Pulse Decoder KY-384/APX-7 and the Control-Indicator C-3707/APX-7"

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*Page 9, paragraph 3.4.2, third sentence: delete and substitute:

"The control labeled 'GATE AMPL' shall be used to adjust the average amplitude of the range-azimuth gate output pulses over the range of 1.0 to 3.0 volts."

Page 9, after paragraph 3.4.3: Add:

"3.4.4 Mounting MT-2648(*)/APX-7 - The mounting shall provide mounting provisions for Pulse Decoder KY-384/APX-7."

*Page 9, paragraph 3.5 b.: Add:

"or Reliability Assurance Tests"

*Page 13, paragraph 4.4.2, line 3: Add

"(When reliability assurance tests of 4.4.3 are conducted, sampling tests shall be omitted)"

*Page 13, after paragraph 4.4.2.1: Add:

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure II from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be Level III from Table I of Specification MIL-R-23094.

4.4.3.3 Accept-Reject Criteria - Figure 12 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 4 hours during each cycle.

4.4.3.5 Performance Characteristics to be Measured -

Characteristic	Spec Requirement	Time in cycle that measurement is to be made
Video Outputs	3.3.3 - 3.3.3.1	One (1) hour after high-low temperature extremes are stabilized.
Decoding	3.3.4 - 3.3.4.4	One (1) hour after high-low temperature extremes are stabilized.
Display	3.3.5 - 3.3.5.2	One (1) hour after high-low temperature extremes are stabilized.

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4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

*Pages 13 and 14: renumber the following paragraphs:

From	To:
4.4.3	4.4.4
4.4.3.1	4.4.4.1
4.4.3.2	4.4.4.2
4.4.4	4.4.5

*Page 18, paragraph 6.4: Add

"(4) Reordered Equipment - The following specifications shall take precedence over those concerning performance requirements listed or referenced in 2.1 where equipment is reordered except in those areas where the contract specifies redesign or modification related to performance.

MIL-I-6181B 29 May 1953"

An asterisk preceding a page number denotes a change to be made to the specification which did not appear in the previous amendment.

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